A
PROJECT REPORT
on

"IOT Based Solar Crop Dryer Monitoring System"

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Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur for Partial Fulfillment of the Degree of

Bachelor of Technology

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CERTIFICATE

This is to certify that project work described in this report entitled, "**IOT BASED SOLAR CROP DRYER MONITORING SYSTEM**" was carried out by **Mr. Ram Dhote, Mr. Aniket Kadu, Mr. Aditya Korde, Mr. Sahil Dethe** in Tulsiramji Gaikwad-Patil College of Engineering & Technology, Nagpur under my supervision and guidance in partial fulfillment of the requirement for the degree of **B.Tech CSE-Data Science** of Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur.

This work is the own work of the candidate, completed in all respects and is of sufficiently high standard to warrant its submission to the said degree. The assistance and resources used for this work are duly acknowledged.

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DECLARATION

I hereby declare that this Project titled **"IOT BASED SOLAR CROP DRYER MONITORING SYSTEM"** is a bonafide and authentic record of the work done by me under supervision of **Dr. Abhimanyu Dutonde** during academic session 2026-27.

The work presented here is not duplicated from any other source and also not submitted earlier for any other degree/diploma of any university. I understand that any such duplication is liable to be punished in accordance with the university rules. The source material, data used in this research have been duly acknowledged.

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CONTENTS

Certificate i
Declaration ii
Acknowledgement iii
Plagiarism Report on the present work iv
Certificate(s) for Project Work ..... v-ix
Contents ..... x-xiii
Table Index xiv
Figure Index xv
Abstract xvi

CHAPTER – I

Overview of Present Work 1-4
1.1 Introduction 1
1.2 Problem Statement 2
1.3 Purpose of the study 3
1.4 Scope and Objective 4

CHAPTER – II

Literature Review And Analysis 5-8
2.1 Evolution of IOT based solar drying system 5
2.2 Research Findings and Trends 6
2.3 Research Findings and Trends 8

CHAPTER – III

SYSTEM REQUIREMENTS AND DESIGN PRINCIPLES 9-13
3.1 System Requirements 9
3.2 Software Requirement 10
3.3 Functional Requirements 11
3.4 Quality Requirements 12
3.5 Software Architecture 13

CHAPTER – IV

TOOLS AND TECHNOLOGIES 14-22
4.1 Design Tools 14
4.1.1 Arduino IDE 15
4.1.2 MQTT 16
4.1.3 MQTT Broker 17

4.1.4 Node-RED ……..17
4.1.5 Influx DB ……..…….18
4.1.6 Grafana ……...19
4.1.7 Local Computer-Based Software Environment …….19
4.1.8 Software Integration ……...20
4.1.9 Python ……21
4.1.10 Visual Studio Code ……..22

CHAPTER – V

SYSTEM DESIGN AND IMPLEMENTATION 23-39
5.1 Requirement analysis and gathering 23
5.1.1 Determining System Needs 24
5.1.2 Identification of Monitoring Parameters 24
5.2 System Architecture and Workflow 26
5.2.1 Sensor Data Acquisition 27
5.2.2 Communication Architecture 27
5.2.3 Sensor Data Processing 28
5.2.4 Sensor Data Storage 29
5.2.5 Data Visualization and Monitoring 30
5.3 Software Design and User Interface 31
5.3.1 System Dashboard Design 32
5.3.2 Real-Time Monitoring 33
5.3.3 Drying Status and Progress 34
5.3.4 Historical Data Visualization 34
5.4 Integration and Implementation 35
5.4.1 ESP32 and Sensor Integration 36
5.4.2 MQTT and Local Computer Integration 36
5.4.3 Node-RED and Database Integration……..37
5.4.4 Grafana Integration ……38
5.4.5 System Integration and Testing ……39

CHAPTER – VI

RESULT AND DISCUSSION 40-43
6.1 Introduction 40
6.2 Implementation Results 40
6.2.1 Data Acquisition and communication 41
6.2.2 Data processing and storage 41

6.2.3 Dashboard Visualization 41
6.3 Functional Verification 41
6.4 Discussion 42
6.5 Achievement of objectives 42
6.6 Limitations 43
6.7 Summary 43

CHAPTER – VII

CONCLUSION AND SUGGESTED FURTHER WORK 44-45
7.1 Conclusion 44
7.2 Suggested Further Work 45

REFERENCES 46-47
Research Paper / Publication .....

TABLE INDEX

Sr. No.	Table No.	Name of Table	Page No.
1	Table 4.1	Arduino IDF and ESP32 Development	15
2	Table 5.1	Identification of monitoring parameters	25
3	Table 5.2	Dashboard Parameters and Visualization	32
4	Table 6.1	Summary of implementation results	40
5	Table 6.2	Functional verification of the system	41
6	Table 6.3	Objectives and outcomes	42

FIGURE INDEX

Sr. No.	Figure No.	Name of Figure	Page No.
1	Fig. 5.1	Shows Node-RED communication	28
2	Fig. 5.2	Shows Sensor data storage in InfluxDB	29
3	Fig. 5.3	Shows Grafana Dashboard visualization	31

ABSTRACT

The integration of IoT, data analytics, and intelligent monitoring has become increasingly Important in the development of modern agricultural systems. Software plays a crucial role in Enabling real-time monitoring, data communication, automated control, and decision-making in smart crop-drying applications. This thesis focuses on the on the software development of a Smart Portable Multi-Crop Drayer, designed to monitor and manage the drying process using sensor-based data and an IoT-enabled architecture. The proposed software system collects temperature, humidity, and weight-related data from the drying chamber and processes this information to provide continuous monitoring of crop-drying conditions. The proposed system integrates an ESP32-based sensing unit with a Raspberry Pi for data provided. Processing and communication. MQTT is used for reliable transmission of sensor data, while Node-RED provides data processing and control-flow management. Influx DB is used for storing time-series data, and Grafana provides an interactive dashboard for real-time visualization of temperature, humidity, weight, drying status, and drying progress. The software architecture is designed to support automated monitoring and control, reducing the need for continuous manual supervision and providing users with accessible information about drying process. The proposed software are solutions aims to improves the efficiency, reliability, and transparency of crop drying by combining real-time sensing, data visualizations, and automated decision making. Continuous monitoring can help maintain suitable drying conditions and reduce the possibility of over-drying or under-drying condition. The system also provides a foundation for future to integration of drying-time prediction, moisture forecasting, data analytics, remote monitoring, and intelligent control techniques. The study demonstrates how IoT and data driven software technologies can contribute to the development of a more efficient, sustainable, and user oriented agricultural drying system.

Keywords: Internet of Things (IoT), Smart Agriculture, Crop Drying, Smart Portable Multi-Crop Dryer, ESP32, Raspberry Pi, MQTT, Node-RED, InfluxDB, Grafana, Real-Time Monitoring, Sensor Data, Temperature Monitoring, Humidity Monitoring, Weight Monitoring, Data Visualization, Automated Control, Time-Series Data, Intelligent Monitoring, Data Analytics.

CHAPTER – I

OVERVIEW OF PRESENT WORK

CHAPTER - I

OVERVIEW OF PRESENT WORK

1.1 Introduction

In modern agriculture, efficient and controlled crop drying plays and important role in Maintaining the quality, safety, and storage life of agricultural products, Traditional drying methods often depend on environmental conditions and manual monitoring, which can result in uneven drying, excessive drying time, or insufficient moisture removal. These limitations create a need for smart and automated drying systems that can continuously monitor drying conditions and assist users in making appropriate decisions during the drying process. The proposed project, IOT based solar crop dryer with monitoring system, integrates IoT and software technologies to provide real-time monitoring and intelligent management of the crop-drying process. The software system collects information such as temperature, humidity, and weight from sensors installed within the drying chambers. An ESP32 is used for sensor data acquisition and communication, while a Local Computer acts as the central computing unit, MQTT enables communication between the sensing and processing units, while Node-RED is used for data processing and system workflow management.

The collected data is stored using the influxDB and presented through a Grafana-based dashboard, allowing users to observe the drying conditions and progress in real time. The software is designed to reduce continuous manual supervision and provides a clear understanding of the drying process through graphical visualization and automated monitoring. The system also provides a foundation for future implementation of drying-time prediction, moisture forecasting, remote monitoring, and intelligent control, there by contributing towards a more efficient, reliable, and data-driven agricultural drying solution.

The systems is designed with portability and ease of uses in mind, making it suitable for different types of crops and small-scale use agricultural applications. By continuously observing all important drying parameters, the system to helps user understand how environmental conditions affect the drying process. This can help reduces unnecessary drying and minimizes the chances of crop damage caused by improper drying conditions. The stored data can also be useful for analyzing previous drying cycles and improving the process in the future. Overall, the project combines the sensing, communication, data storage, and provides visualization technologies to create a practical solution for modern crop drying. It aims to make the drying process more systematic, less dependent on manual observation, and easier to monitor and manage.

1.2 Problem Statement

Traditional crop-drying methods Often depend heavily on environmental conditions and manual supervision, making it difficult to maintain consistent and stable drying conditions. Variations it's temperature, humidity, and crop moisture can lead to uneven drying, over-drying, or insufficient moisture removal, which may affect the quality of storage life of agricultural products. In addition, conventional methods generally lack real-time monitoring and automated detection decision making capabilities.

The absence of an integrated software system for continuously collecting, processing, storing, and visualizing drying data makes it difficult for users to accurately track the progress of the drying process. Therefore, there is a need for an IoT-based software solution that can acquire sensor data provide real-time monitoring and visualization of drying conditions, and support automated control and future prediction of the crop-drying process.

Traditional crop drying methods often depend on natural environmental conditions such as sunlight, temperature, humidity, and airflow. Since these conditions can change throughout the day, it becomes difficult to maintain a stable and controlled drying environment. Farmers and workers also need to regularly check the crops manually, which can be time-consuming and may not always provide accurate information about the actual drying condition. Changes in temperature, humidity, and crop moisture can result in uneven drying. Some portions of the crop may remain wet while others may become excessively dry. This can affect the quality storage life, and overall value of agricultural products.

In addition, conventional drying methods generally do not provide real-time information about the drying process, making it difficult to identify problems at the right time. Another major limitation is the lack of an integrated system for collecting, processing, storing, and visualizing drying-related data. Users may not have a clear record of how temperature, humidity, and weight change during the drying process. Therefore, there is a need for an IoT-based software solution that can continuously collect sensor data and display it in an easy-to-understand form. Such a system can help reduce manual monitoring, improve process understanding, and provide a foundation for automated control, data analysis, and future prediction of drying time and moisture levels.

1.3 Purpose of the study

The purpose is to develop a **IOT based solar crop dryer with monitoring system** with the following objectives:

1. **Enable real-time monitoring** of temperature**,** humidity, and weight-related parameters during the crop-drying process.
2. **Collect and transmit sensor data** using an ESP32 and MQTT communication for continuous monitoring of the drying chamber.
3. **Process and manage the collected data** using Local Computer and Node-RED to support efficient monitoring and control of the drying operation.
4. **Store historical drying data** using influxDB to maintain records that can be used for analysis and performance evaluation.
5. **Provide an interactive dashboard** using Grafana to display temperature, humidity, weight, drying status, and drying progress in an understandable format.
6. **Reduce manual supervision** by providing automated monitoring and supporting intelligent decision-making during the drying process.
7. **Provides a foundation for future features** such as drying-time prediction, moisture forecasting, remote monitoring, and data-driven optimization of the drying process.

The main purpose is to develop a IOT based solar crop dryer with monitoring system. The system is intended to make the crop-drying process easier to monitor, more organized, and less dependent on continuous manual supervision. It focuses on collecting useful information from the drying chamber and presenting it to the user in a simple and understandable way.

The major objectives of the project are to continuously monitor parameters such as temperature, humidity, and crop weight during the drying process. An ESP32 is used to collect sensor readings and transmit them through MQTT**.** The Local Computer and Node-RED are used to receive, process, and manage the collected data. Historical data is stored in InfluxDB**,** which helps in reviewing and analyzing previous drying cycles. A Grafana dashboard is developed to visualize important parameters and drying progress in real time. The system also aims to reduce manual monitoring and provides a foundation for futures features such as drying-time predictions, moisture forecasting, remote monitoring, and data-driven optimization.

1.4 Scope and Objective

Scope:

1. **Creating the software architecture** for the IOT based solar crop dryer with monitoring system's real-time administration and monitoring.
2. **Using the ESP32-based sensing** system to gather sensor data from the drying chamber, including weight, temperature, and humidity.

3. **Continuous monitoring** by sending sensor data between the ESP32 and Local Computer using Wi-Fi and MQTT.
4. **Utilizing Node-RED** for data processing and management to facilitate automated decision-making, control logic, and monitoring.
5. **Keeping past sensor data in InfluxDB** for drying process analysis, comparison, and assessment.
6. **Creating an interactive Grafana dashboard** that shows temperature, humidity, weight, drying status, and drying progress in real time.
7. **Encouraging automated drying process** monitoring and control using the sensor data gathered.
8. **Providing a foundation for future enhancements,** such as drying-time prediction, moisture forecasting, remote monitoring and intelligent data driven drying control.

The first objective of the proposed software system is to enable real-time monitoring of the crop drying process. The system continuously collects parameters such as temperature, humidity, and weight from the drying chamber. This information allows the user to observe the current drying conditions and understand how the process is progressing without depending entirely on manual observation.

Another important objective is reliable data communication and processing. The ESP32 collects sensor readings and transmits the data through Wi-Fi using the MQTT protocol. The Local Computer receives and processes the information while Node-RED manages the data flow and control logic. This software architecture enables continuous communication between sensing and monitoring components. The system also aims to store and visualize drying data effectively**.** InfluxDB is used to store time-series sensor data while Grafana provides an interactive dashboard for displaying temperature, humidity, weight, drying status, and drying progress. Historical data can also be used to analyze different drying cycles and understand the behavior of the drying process.

Another objective is to reduce manual supervision through automated monitoring and control. By continuously analyzing sensor readings, the software can identify changes in drying conditions and support appropriate control decisions. This can help maintain stable suitable operating conditions and reduce the chance of over-drying or under-drying.

Finally, the software is designed with future scalability in mind. The system can be extended with features such as drying time prediction, moisture forecasting, remote monitoring, machine-learning-based analysis, and intelligence optimization. This makes the proposed software architecture suitable for further development and adoption for different crops and drying conditions.

CHAPTER – II

LITERATURE REVIEW AND ANALYSIS

CHAPTER - II

LITERATURE REVIEW AND ANALYSIS

2.1 Evolution of IOT based solar drying system

Traditional open sun drying is a low-cost method for preserving agricultural products but it's highly dependent on weather conditions and requires continuous manual supervision. Variations in temperature, humidity, contamination, and drying duration can affect the quality and consistency of the final product. Recent research has therefore focused on integrating solar drying systems with Internet of Things (IoT) technologies to enable real-time monitoring, data collection, and automated control of drying conditions.

Damayanti et al. developed a dual-energy solar drying system that utilizes solar energy under suitable weather conditions and an interactive alternative energy source during cloudy periods or at night. The system incorporated IoT-based monitoring of temperature and humidity, allowing users to observe drying conditions and maintain more stable operating conditions.

Hasan et al. Modified a solar dryer by integrating temperature, humidity, and light sensors for monitoring the drying environment. The collected parameters were transmitted to a control unit, allowing users to monitor and control the drying process through an interface. The study demonstrated that sensor-based monitoring could improve the consistency and efficiency of drying compared with conventional sun drying.

Latif et al. proposed an IoT-based grain-drying framework using temperature and humidity sensors along with an HX711 load-cell system. The system continuously monitored product weight and environmental conditions. The collected data was used to estimate moisture loss and support closed-loop drying control, demonstrating the importance of combining sensor data with automated decision-making.

Lopez-Velasco al. developed an IoT monitoring system for a solar greenhouse dryer in which temperature and relative humidity data were transmitted through LolRa and Wi-Fi communication technologies to a web-based monitoring platform. The system also incorporated a predictive model for estimating banana moisture content, demonstrating the potential of combining IoT monitoring with predictive analytics.

2.2 Research Findings and Trends

The review studies indicate that IoT-based drying systems are gradually developing from simple environmental monitoring systems towards an integrated platform capable of monitoring,

prediction, and automated temperature control. Temperature and humidity sensors are commonly used to observe drying conditions while a load cell can be used to monitor changes in product mass and estimate moisture loss. These sensor-based approaches provide continuous information that can be used to evaluate the drying process more effectively.

Raj et al. Developed a low-cost solar cabinet dryer using an ESP32 controller, temperature and humidity sensors, and a load cell for measuring product mass. The system provided remote monitoring and demonstrated the application of feedback-based control for managing the drying conditions. Such systems show that low-cost microcontrollers can be effectively integrated with sensors and communication technologies for agricultural applications. The review studies also demonstrate the importance of real-time dashboard and user interface platforms such as Blynk, ThingSpeak, and web-based interface allows users to observe drying parameters remotely and make timely operational decisions. This reduces the dependency on continuous manual supervision and improves the visibility of the drying process. The reviewed studies show that IoT technology is becoming increasingly useful in agricultural drying systems. Earlier drying systems mainly focused on measuring environmental conditions, whereas recent systems are moving towards more connected solutions that can monitor, analyze, and control the drying process. Parameters such as temperature and humidity are commonly monitored because they directly influence the drying rate and quality of agricultural products. In addition, load cells can be used to measure changes in crop weight during drying. The decrease in weight can provide useful information about moisture loss and help in understanding the progress of the drying process.

Raj et al. developed a low-cost solar cabinet dryer using an ESP32 controller along with temperature and humidity sensors and a load cell. Their system demonstrated how sensor data can be collected and monitored remotely while also supporting feedback-based control of drying conditions. This study highlights that affordable microcontrollers and sensors can be combined to develop practical agricultural solutions without requiring highly expensive equipment.

The reviewed systems also highlight the importance of real-time dashboards and user-friendly interfaces. Platforms such as Blynk, ThingSpeak, and web-based dashboards allow users to view sensor readings remotely and understand the current condition of the drying chamber. Graphical representation of temperature, humidity, and weight makes the information easier to interpret and helps users take timely decisions. Overall, these studies provide a useful foundation for developing the proposed IOT based solar crop dryer with monitoring system, particularly in areas such as sensor integration, real-time monitoring, data storage, visualization, and future automation.

2.3 Identified research Gap

There are still a number of chances to create a more integrated software architecture, even if previous research has effectively shown IoT-based monitoring, sensor integration, remote visualization, and automated control in solar and agricultural drying systems. Few systems integrate real-time data acquisition, IoT communication, data processing, historical data storage, visualization, and intelligent decision support into a single software framework, whereas many current systems concentrate on specific tasks like sensor monitoring or remote visualization.

Latif et al. found that the combination of automated control and sensor monitoring reduced drying time and increased energy efficiency. In a similar vein, studies examined by Tiwari et al. show that solar-drying technology has advanced from direct and indirect dryers to hybrid systems backed by microcontrollers and Internet of Things sensors. Nonetheless, there is still much room for improvement in the integration of automated control, remote visualization, energy-related data, predictive analytics, and continuous real-time monitoring.

Thus, the integration of sensor data acquisition via ESP32, MQTT-based communication, Node-RED workflow management, InfluxDB for time-series data storage, and Grafana for real-time visualization is the main goal of the proposed software component of the IOT based solar crop dryer with monitoring system. Future features like moisture forecasting, drying-time prediction, remote monitoring, and intelligent drying control are made possible by this integrated approach.

Although previous studies have demonstrated the usefulness of IoT technology in agricultural and solar drying systems, there is still scope for developing a more complete and integrated software architecture. Many existing systems focus mainly on individual functions such as collecting sensor readings, monitoring temperature and humidity, or displaying data remotely. These functions are useful, but when they are not connected within a single software framework, it can become difficult to manage the complete drying process and analyze the collected information effectively.

Latif et al. reported that combining sensor monitoring with automated control can help reduce drying time and improve energy efficiency. Similarly, the studies reviewed by Tiwari et al. indicate that solar drying systems have gradually developed from basic direct and indirect dryers towards hybrid systems that use microcontrollers and IoT-based sensors. However, opportunities still exist to combine real-time monitoring, automated control, remote visualization, energy-related information, and predictive analysis more effectively.

To address this gap, the proposed IOT based solar crop dryer with monitoring system focuses on connecting different software and hardware components into one integrated system. The ESP32 collects sensor data, while MQTT provides communication between the devices. A Local Computer acts as the central processing unit, and Node-RED manages data processing and system

workflows. InfluxDB stores time-series data, while Grafana provides real-time graphical visualization. This architecture creates a strong foundation for future features such as moisture prediction, drying-time estimation, remote monitoring, and intelligent control.

CHAPTER – III

SYSTEM REQUIREMENTS AND DESIGN PRINCIPLES

CHAPTER - III

SYSTEM REQUIREMENTS AND DESIGN PRINCIPLES

3.1 System Requirements

The software environment needed for the proposed IOT based solar crop dryer with monitoring system must be able to collect sensor data, transport it, process it, keep track of its history, and show the user the drying conditions. A Local Computer with Arduino IDE-based processing environment and an ESP32-based sensing unit form the foundation of the software architecture. The system offers a centralized platform for drying process monitoring and transfers sensor information via wireless connection. The drying chamber's temperature, humidity, and crop weight are all recorded by the sensing device. These readings are obtained and sent via Wi-Fi by the ESP32. Because MQTT is lightweight and appropriate for Internet of Things applications, it serves as the communication protocol between the sensing unit and the processing system.

The computational infrastructure needed to process and manage the sensor data is provided by the Raspberry Pi. The data flow is arranged and the necessary monitoring and control logic is implemented using Node-RED. InfluxDB, which can handle continually generated time-series sensor data, is used to store the gathered data. The processed data is shown via a graphical dashboard using Grafana, which enables the user to monitor the drying conditions and advancement. The software environment for the proposed IOT based solar crop dryer with monitoring system is designed to make the overall drying process easier to monitor, manage, and analyze. The system combines different hardware and software components so that sensor readings can be collected, transferred, processed, stored, and displayed in a structured manner. The main purpose of this environment is to provide users with clear information about the drying conditions without requiring them to continuously check the drying chamber manually.

The ESP32 works as the sensing and communication unit of the system. Sensors placed inside the drying chamber measure important parameters such as temperature, humidity, and crop weight. The ESP32 collects these readings at regular intervals and uses Wi-Fi to send the data to the processing system. MQTT is used as the communication protocol because it is lightweight, reliable, and well suited for IoT-based applications. It allows sensor data to be transferred efficiently between different components of the system.

A Local Computer provides the main computing environment for processing and managing the received sensor information. Node-RED is used to create the data flow and manage different

processing and monitoring operations. The collected readings are stored in InfluxDB, which is suitable for handling continuously generated time-series data. Finally, Grafana converts the stored data into interactive graphs and dashboards. Through these visualizations, users can easily observe changes in temperature, humidity, and crop weight and understand the progress of the drying process. This integrated environment also provides a foundation for future automation, prediction, and remote monitoring features.

3.2 Software Requirement

Python is needed for any further data processing or supporting scripts, and the Arduino IDE is needed for programming the ESP32 in the software development of the proposed system. MQTT, Node-RED, InfluxDB, and Grafana are used by the Raspberry Pi, which serves as the primary software platform, to set up the entire monitoring pipeline.

1. The following are the main software technologies employed in the suggested system: The ESP32 firmware is developed and uploaded using the Arduino IDE.
2. MQTT: This protocol allows the ESP32 and Local Computer to exchange sensor data.
3. MQTT Broker: Oversees the communication between the processing and sensing devices.
4. Node-RED: Used for processing, automation logic, and data flow management.
5. InfluxDB: Used to keep track of past drying data and store sensor readings.
6. Grafana: This tool is used to generate the monitoring dashboard and display both historical.

The proposed IOT based solar crop dryer with monitoring system uses a combination of software technologies to create a complete monitoring and data-management environment. Each technology has a specific role in the system, and together they help in collecting sensor readings, transferring the data, processing it, storing historical information, and presenting the results to the user. The software architecture is designed to be simple, flexible, and suitable for future improvements.

The Arduino IDE is used to develop and upload the firmware to the ESP32. The ESP32 reads values from sensors such as temperature, humidity, and weight sensors and prepares the data for transmission. MQTT is used as the communication protocol between the ESP32 and Raspberry Pi. Since MQTT requires low communication overhead, it is suitable for continuous sensor-data transmission in IoT applications.

An MQTT Broker manages the exchange of messages between the sensing and processing devices. It receives the sensor information published by the ESP32 and makes it available to the required processing components. On the Raspberry Pi, Node-RED is used to create the data flow and handle processing, automation logic, and communication between different software components. The processed sensor readings are stored in InfluxDB, which is designed for time-series data and allows

previous drying records to be maintained for later analysis. Grafana is then connected to InfluxDB to create an interactive dashboard where current and historical readings can be viewed through graphs and other visual elements. The Local Computer OS provides the operating environment for these software components. In addition, Python can be used for supporting scripts, data analysis, calculations, and future features such as drying-time prediction and moisture estimation. Overall, these technologies work together to provide a structured and scalable software platform for the proposed dryer.

3.3 Functional Requirements

It is anticipated that the suggested software system will continuously gather and process data produced during the drying process. Using Wi-Fi and MQTT, the system should gather sensor values via the ESP32 and send them to the Raspberry Pi. After then, the data should be processed and saved for later use.

Important drying factors like temperature, humidity, crop weight, drying status, and drying progress should be visible to the user through the system's graphical user interface. In order to analyze the drying process after it is finished, it needs also save historical records. Additionally, automated drying condition monitoring should be supported by the software. The system can supply the data required to regulate and oversee the drying process based on the sensor data gathered. Future development should be able to incorporate more analytical and prediction features thanks to the architecture.

The proposed software system is expected to provide continuous monitoring and management of the crop-drying process. The main purpose of the system is to collect useful information from the drying chamber and make it available to the user in a simple and understandable form. The ESP32 collects sensor readings such as temperature, humidity, and crop weight at regular intervals. These readings are transmitted through Wi-Fi using the MQTT communication protocol to the Local Computer for further processing.

After receiving the data, the Local Computer processes the sensor information and manages the flow of data through Node-RED. The processed readings are stored in InfluxDB so that the information can be accessed later for analysis and comparison between different drying cycles. Maintaining historical records is useful because it allows users to understand how the drying conditions changed over time and evaluate the performance of the dryer. The system is also expected to provide a graphical interface through Grafana. The dashboard will display important parameters such as current temperature, humidity, crop weight, drying status, and overall drying progress. Presenting this information through graphs and visual indicators can make it easier for users to understand the condition of the crops without checking the sensors manually.

Another important function of the proposed system is continuous monitoring of drying conditions. Based on the collected sensor data, the system can provide information that may support appropriate control actions during the drying process. The software architecture is also designed with future development in mind. Additional features such as drying-time prediction, moisture-level estimation, alerts, remote monitoring, and intelligent control can be added later as more data becomes available. This makes the system flexible and suitable for further improvement.

3.4 Quality Requirements

Reliability, usability, scalability, and maintainability should all be features of the software system. For the monitoring data to be reliable throughout the drying cycle, sensor data should be sent and recorded consistently. To make it easy for customers to interpret the dryer's current state, the dashboard should display information in an easy-to-understand format. Additionally, the software architecture should be modular, enabling the expansion or modification of specific elements like processing, data storage, communication, and display without having to completely redesign the system. This is especially crucial for the future integration of features like intelligent data analysis, moisture forecasting, drying-time prediction, and remote monitoring.

The software system of the proposed IOT based solar crop dryer with monitoring system should not only perform its basic monitoring functions but should also be reliable, easy to use, scalable, and simple to maintain. Since the system is expected to operate continuously during the drying process, reliability is an important requirement. Sensor readings should be collected and transmitted consistently so that users can depend on the displayed information while making decisions about the drying process.

Usability is another important aspect of the system. The dashboard should present information in a clear and understandable manner so that users do not need advanced technical knowledge to understand the current condition of the dryer. Important parameters such as temperature, humidity, crop weight, and drying progress should be displayed using suitable graphs, values, and indicators. A simple interface can make the system more convenient for regular use.

The system should also be scalable so that additional sensors, devices, or software features can be added when required. For example, new parameters such as moisture level, energy consumption, or airflow can be incorporated in the future without affecting the complete system.

Maintainability is also considered while designing the software architecture. The system should follow a modular structure where communication, data processing, storage, and visualization are handled separately. This makes it easier to identify problems, update individual components, and improve the system over time. Such an architecture will also support the future addition of features

such as intelligent data analysis, moisture forecasting, drying-time prediction, automated alerts, and remote monitoring.

3.5 Software Architecture

A sequential data-processing architecture underpins the entire program function. The ESP32 initially gathers sensor values, which are then sent over Wi-Fi using MQTT. Node-RED controls the data flow while the Local Computer receives the information. After processing, the readings are saved in InfluxDB and made accessible to Grafana for both historical and real-time display. The software architecture of the proposed IOT based solar crop dryer with monitoring system is designed as a sequential and connected data-processing system. The main purpose of this architecture is to collect information from the drying chamber, transfer it to the processing unit, store the data safely, and present useful information to the user. Each component performs a specific task, and the output from one stage becomes the input for the next stage.

The process begins with the ESP32, which acts as the sensing and data-acquisition unit. Sensors installed inside the drying chamber continuously measure important parameters such as temperature, humidity, and crop weight. The ESP32 collects these readings at regular intervals and prepares them for transmission. Using its built-in Wi-Fi capability, the ESP32 sends the collected sensor data to the Local Computer through the MQTT communication protocol. An MQTT broker manages the exchange of messages between the ESP32 and the Raspberry Pi. Once the data reaches the Raspberry Pi, Node-RED is used to manage the complete data flow. It receives the sensor readings, processes the information, and passes the required data to the appropriate services. This makes it easier to organize the monitoring workflow and provides flexibility for adding control or automation logic in the future.

After processing, the sensor readings are stored in InfluxDB. Since drying data is generated continuously over time, InfluxDB is suitable for maintaining time-series records. These historical records can later be used to study previous drying cycles and compare changes in temperature, humidity, and crop weight.

Finally, Grafana is connected to the stored data and provides a user-friendly dashboard. The dashboard can display both real-time and historical information using graphs, values, and other visual elements. This allows users to understand the current drying conditions and observe the progress of the drying process without continuously checking the physical sensors. Overall, the architecture provides a structured foundation for future features such as automated control, drying-time prediction, moisture forecasting, alerts, and remote monitoring.

CHAPTER – IV

TOOLS AND TECHNOLOGIES

CHAPTER - IV

TOOLS AND TECHNOLOGIES

4.1 Design Tools

To establish connection between the sensing unit, processing system, database, and user interface, a set of development and monitoring tools is needed for the implementation of the suggested IOT based solar crop dryer with monitoring system software. The tools chosen are determined by how well they work with IoT data collection, processing, storage, and real-time display. To establish connection between the sensing unit, processing system, database, and user interface, a set of development and monitoring tools is needed for the implementation of the suggested IOT based solar crop dryer with monitoring system software. The tools chosen are determined by how well they work with IoT data collection, processing, storage, and real-time display.

Arduino IDE is used for developing and uploading the program to the ESP32. It helps in reading the connected temperature, humidity, and crop weight sensors and preparing their values for transmission. The ESP32 uses Wi-Fi to communicate with the rest of the system, while MQTT is used as the communication protocol for transferring sensor data.

The Local Computer provides the main processing environment for the software components. Node-RED is used to create and manage the data flow between the MQTT broker, database, and dashboard. Its visual flow-based approach makes it easier to handle incoming sensor information and connect different services without making the overall system unnecessarily complex.

InfluxDB is selected for storing the sensor readings because the data generated by the dryer is mainly time-based. It keeps the measurements along with timestamps, making it suitable for maintaining historical records of the drying process. Grafana is connected to InfluxDB to present this information through graphs, numerical values, and other dashboard elements.

Python can also be used for supporting scripts, calculations, and future data analysis. Together, these tools create a connected software environment in which data can move from sensors to communication, processing, storage, and finally to the user interface. The modular nature of these tools also makes the system easier to maintain and expand with future features such as alerts, prediction, and intelligent drying control.

4.1.1 Arduino IDE

The firmware is developed and uploaded to the ESP32 DevKit using the Arduino IDE. The software is in charge of establishing Wi-Fi connectivity, reading linked sensor readings, and getting the gathered data ready for MQTT delivery.

Table 4.1 Arduino IDE and ESP32 Development

Sr.no	Parameter	Description
1	Development Tool	Arduino IDE
2	Target Device	ESP32 DevKit
3	Programming language	C/C++ based Arduino programming
4	Main Function	Sensor acquisition and communication
5	Connectivity	Wi-Fi
6	Sensor parameters	Temperature, humidity, crop weight
7	Communication Protocol	MQTT
8	Data Preparation	Sensor readings are formatted before transmission
9	Debugging method	Serial Monitor
10	Firmware management	Compile, upload, modify and update ESP32 firmware

The firmware is developed and uploaded to the ESP32 DevKit using the Arduino IDE. The software is in charge of establishing Wi-Fi connectivity, reading linked sensor readings, and getting the gathered data ready for MQTT delivery. Arduino IDE provides a simple development environment for writing, compiling, and uploading the program to the ESP32.

The firmware is designed to communicate with the connected sensors and collect important drying parameters such as temperature, humidity, and crop weight. The ESP32 reads these values at regular intervals and organizes the readings so that they can be transferred to the next stage of the monitoring system. This allows the sensing unit to continuously provide updated information about the drying environment.

The Wi-Fi functionality of the ESP32 is also configured through the firmware. Once the connection is established, the device can communicate with the MQTT broker over the network. The collected sensor readings are prepared in a suitable format and published through the required MQTT topics. This creates a continuous communication path between the sensing unit and the Raspberry Pi-based processing system.

The Arduino IDE also makes it easier to modify and update the firmware whenever changes are required. For example, sensor reading intervals, communication settings, or additional sensor-related functions can be adjusted during development and testing. Serial monitoring can also be used during development to observe sensor readings and identify basic communication issues.

Overall, Arduino IDE provides the development environment required to program the ESP32 and manage the sensing and communication tasks. It forms the starting point of the software pipeline, where physical sensor measurements are converted into digital data that can be transmitted, processed, stored, and visualized by the remaining components of the IOT based solar crop dryer with monitoring system.

4.1.2 MQTT

The ESP32 and the Local Computer environment communicate using MQTT (Message Queuing Telemetry Transport). It makes it possible to send the sensor data effectively utilizing a publish-subscribe communication mechanism.

The ESP32 and the Local Computer environment communicate using MQTT (Message Queuing Telemetry Transport). It makes it possible to send sensor data effectively by using a publish-subscribe communication mechanism. MQTT is a lightweight communication protocol commonly used in IoT systems where devices need to exchange data efficiently over a network. In the proposed IOT based solar crop dryer with monitoring system, MQTT acts as an important communication layer between the ESP32 and the Raspberry Pi.

The ESP32 collects temperature, humidity, and crop weight readings from the connected sensors. These readings are organized into messages and published through MQTT over the Wi-Fi network. An MQTT broker manages these messages and forwards them to the required subscriber. The Local Computer environment receives the published data and passes it to Node-RED for further processing and storage. This reduces the need for direct communication between the ESP32 and individual software components.

The publish-subscribe approach also makes the system flexible because different applications can subscribe to the same sensor data when required. MQTT requires relatively low network resources, making it suitable for continuous sensor monitoring. It also helps maintain a clear separation between data collection and data processing. The communication system can be extended in the future to support additional sensors, alerts, remote monitoring, and automated drying-control functions.

4.1.3 MQTT Broker

The ESP32 and the receiving software components exchange messages under the control of the MQTT Broker. After receiving the published sensor readings, it makes them accessible to the relevant subscribers so they can be processed further.

The ESP32 and the receiving software components exchange messages under the control of the MQTT Broker. After receiving the published sensor readings, it makes them accessible to the relevant subscribers so they can be processed further. In the proposed IOT based solar crop dryer with monitoring system, the MQTT Broker works as an intermediate communication point between the ESP32 and the software running on the Raspberry Pi.

The ESP32 collects temperature, humidity, and crop weight data from the connected sensors and publishes the readings to predefined MQTT topics. The MQTT Broker receives these messages and manages their delivery to the appropriate subscribers. On the Raspberry Pi, Node-RED subscribes to the required topics and receives the sensor data for further processing. This arrangement avoids the need for the ESP32 to communicate directly with every software component.

The broker also helps keep the communication structure simple and organized. Since publishers and subscribers do not need to know each other directly, new devices or software services can be added without making major changes to the existing system. This makes the overall architecture more flexible and easier to maintain. The MQTT Broker also supports continuous data exchange, which is useful for real-time monitoring of the drying process. In the future, additional sensors, monitoring services, alert systems, and automated control functions can be connected through suitable MQTT topics.

4.1.4 Node-RED

The software workflow for handling the received sensor data is designed using Node-RED. After receiving data from the MQTT broker, it arranges the readings and directs the processed data to the database and other necessary functions.

The software workflow for handling the received sensor data is designed using Node-RED. After receiving data from the MQTT broker, it arranges the readings and directs the processed data to the database and other necessary functions. In the proposed IOT based solar crop dryer with monitoring system, Node-RED works as an important processing layer between the MQTT communication system, InfluxDB database, and Grafana dashboard.

Node-RED receives the sensor readings published by the ESP32 through the MQTT broker. The incoming data contains information such as temperature, humidity, and crop weight. Node-RED processes and organizes these values so that they can be stored and used by other components of

the system. It can also add timestamps and perform basic checks on the received readings before sending them to InfluxDB for historical storage.

The flow-based structure of Node-RED makes the system easier to understand and manage because different operations can be connected through visual nodes. This reduces the complexity of developing the complete data-processing workflow. It also allows changes to be made to individual parts of the process without affecting the entire system.

Node-RED provides a flexible foundation for future improvements. Additional processing steps can be introduced for generating alerts, identifying unusual readings, monitoring drying progress, or supporting automated control. Therefore, Node-RED helps connect the communication, processing, storage, and visualization components into a continuous software workflow.

4.1.5 InfluxDB

The time-series database used to store the sensor values produced during the drying process is called InfluxDB. In order to analyze both present and historical data, temperature, humidity, crop weight, and relevant timestamps are kept.

The time-series database used to store the sensor values produced during the drying process is called InfluxDB. It is designed to handle data that changes continuously over time, making it suitable for storing sensor readings generated by the IOT based solar crop dryer with monitoring system. Temperature, humidity, crop weight, and relevant timestamps are recorded so that both present and historical data can be used for monitoring and analysis.

The sensor data collected by the ESP32 is transmitted through MQTT and processed using Node-RED before being stored in InfluxDB. Each reading is associated with a timestamp, which helps maintain the sequence of data throughout the drying process. This allows the system to track how the drying conditions and crop weight change over different periods.

InfluxDB provides an organized way to maintain historical records without depending on manual data entry. The stored information can later be accessed by Grafana to create graphs and other visual representations. By observing these trends, the user can understand changes in temperature, humidity, and crop weight during different stages of drying.

The historical data can also be useful for comparing different drying cycles and identifying common patterns. In the future, the stored data can support additional features such as drying-time estimation, moisture prediction, performance analysis, alerts, and data-driven optimization. Thus, InfluxDB forms an important storage layer in the overall monitoring system.

4.1.6 Grafana

The monitoring dashboard is implemented using Grafana. It uses graphs, numerical figures, and status indicators to display the drying parameters after retrieving the stored data from InfluxDB. This gives the user an unobstructed view of the drying process.

The monitoring dashboard is implemented using Grafana. It uses graphs, numerical figures, and status indicators to display the drying parameters after retrieving the stored data from InfluxDB. This gives the user a clear and unobstructed view of the drying process. Grafana acts as the visualization layer of the IOT based solar crop dryer with monitoring system and helps the user understand the current operating conditions without checking individual sensor readings manually.

The data collected by the ESP32 is processed through Node-RED and stored in InfluxDB along with the corresponding timestamps. Grafana connects to this stored data and presents it in an easy-to-understand dashboard. Important parameters such as temperature, humidity, and crop weight can be displayed using numerical values and time-based graphs. These visual representations help the user observe how the drying conditions change over time.

Historical graphs are also useful for reviewing previous drying stages and comparing changes in the monitored parameters. For example, the decrease in crop weight can provide an indication of moisture removal during the drying process. Similarly, temperature and humidity trends help the user understand the environmental conditions inside the dryer.

The dashboard is designed to provide simple and centralized monitoring. In future development, it can be extended with alerts, drying-status indicators, prediction results, remote monitoring, and other intelligent features. Thus, Grafana provides an important interface between the stored sensor data and the user.

4.1.7 Local Computer-Based Software Environment

The central computational environment for the software implementation is provided by the Arduino IDE. This setup enables the primary software components to function as a single monitoring platform, including the MQTT Broker, Node-RED, InfluxDB, and Grafana services.

The central computational environment for the software implementation is provided by the Arduino IDE. It acts as the main processing and hosting platform for the different software services used in the IOT based solar crop dryer monitoring system. Instead of running each software component on separate systems, the Arduino IDE provides a centralized environment where the MQTT Broker, Node-RED, InfluxDB, and Grafana can work together as a single monitoring platform. This makes the overall system more organized and easier to manage.

The Arduino IDE receives sensor data from the ESP32 through the MQTT communication system. The MQTT Broker running in the Arduino IDE environment manages the incoming messages and makes them available to the required software components. Node-RED then receives the sensor readings and performs the required data-processing and workflow operations. After processing, the relevant information is stored in InfluxDB as time-series data along with timestamps.

Grafana is connected to the InfluxDB database and uses the stored information to provide a graphical monitoring interface. Through this arrangement, the complete data flow can be handled within the Arduino IDE environment, starting from receiving sensor information and continuing through processing, storage, and visualization. The user can therefore monitor important drying parameters such as temperature, humidity, and crop weight from a centralized dashboard.

The Arduino IDE environment also supports continuous operation of the monitoring system. It can run the required services together and maintain communication between the different software layers. This reduces the complexity of the overall setup and provides a practical platform for the project's software implementation.

The modular nature of the environment also allows future improvements to be added without changing the complete system architecture. Additional sensors, data-processing functions, alerts, prediction methods, remote monitoring, and intelligent drying-control features can be integrated when required.

4.1.8 Software Integration

The selected tools are integrated to establish a continuous data flow from the sensors to the user dashboard. The complete implementation follows the sequence:

Sensors

↓

ESP32

↓

Wi-Fi

↓

MQTT Broker

↓

Node-RED

↓

InfluxDB

↓

Grafana

↓

User Dashboard

4.1.9 Python

Python can be utilized to support data processing, data analysis, and the drying system's future incorporation of machine learning or predictive features.

Python is used as a supporting programming language for data processing, data analysis, and the future integration of machine learning and predictive features in the IOT based solar crop dryer with monitoring system. Its simple syntax and wide range of libraries make it suitable for working with sensor data collected during the drying process. Python can work alongside the existing IoT software components and provide additional data-processing capabilities when required.

The sensor readings collected by the ESP32 include important parameters such as temperature, humidity, and crop weight. These values can be stored in the InfluxDB database and later used by Python for further analysis. Python can be used to read the stored data, organize it, identify changes in sensor values, and calculate useful information from the recorded readings. This can help in understanding the behaviour of the drying process over different periods.

Python can also support the analysis of historical drying data. For example, changes in crop weight can be examined to understand the progress of moisture removal, while temperature and humidity values can be studied to identify their relationship with the drying process. Such analysis can help in identifying patterns and improving the understanding of different drying conditions.

Another important role of Python is its potential use in future machine learning applications. Once sufficient historical drying data is collected, machine learning models can be developed to estimate drying time, predict moisture levels, identify unusual conditions, or support intelligent decision-making. Python provides several libraries that can be used for these tasks, including NumPy and Pandas for data handling and analysis, along with machine learning libraries such as Scikit-learn.

Python can therefore be considered an additional analytical layer that complements the existing ESP32, MQTT, Raspberry Pi, Node-RED, InfluxDB, and Grafana architecture. Its use provides flexibility for future development and allows the system to move beyond basic monitoring toward data-driven analysis, prediction, and intelligent drying management.

4.1.10 Visual Studio Code

Python scripts, configuration files, Node-RED-related programming, and other software components of the suggested system can be managed using Visual Studio Code as the development environment.

Visual Studio Code is used as the main development environment for managing the software components of the proposed IOT based solar crop dryer with monitoring system. It provides a convenient workspace for writing, editing, organizing, and testing different types of project files. Since the system includes multiple software technologies, Visual Studio Code helps keep the development process structured and manageable.

Python scripts used for data processing and analysis can be created and maintained using Visual Studio Code. The editor provides features such as syntax highlighting, code suggestions, error identification, and an integrated terminal, which make software development easier. These features are useful while developing and testing Python programs that work with sensor data and support future analytical or machine learning functions.

Visual Studio Code can also be used to manage configuration files and other supporting files required by the system. Project files can be organized into separate folders according to their functions, making it easier to locate and modify specific components. This is particularly useful because the proposed system involves communication, data processing, storage, visualization, and analytical components.

During development, the integrated terminal can be used to run Python scripts, install required packages, and perform basic testing. The development environment also helps in identifying programming errors and making necessary corrections before integrating the software into the complete system. This improves the overall development and testing workflow.

Visual Studio Code also provides flexibility for future development of the project. New Python scripts, data-processing functions, machine learning models, configuration files, or additional software modules can be added as the system is expanded. The same environment can therefore support both the current implementation and future improvements.

Overall, Visual Studio Code provides a centralized and user-friendly development environment for managing the software side of the IOT based solar crop dryer with monitoring system. It supports organized coding, testing, debugging, and maintenance, while making it easier to integrate different software components into the overall system.

CHAPTER – V

SYSTEM DESIGN AND IMPLEMENTATION

CHAPTER - V

SYSTEM DESIGN AND IMPLEMENTATION

5.1 Requirement analysis and gathering

The system design process begins with understanding the requirements of the smart portable multi-crop drying dryer and identifying the software functions required for effective monitoring and management of the drying process, the requirements are derived from the operating conditions of the dryer, the parameters that need to be monitored, and the interaction between the hardware and software components. The suggested system is made to keep an eye on crucial variables including crop weight, humidity, and temperature. Continuous sensor-data collection, wireless connectivity, data processing, storing historical data, and real-time visualization are additional requirements. The software architecture is designed utilizing ESP32, MQTT, Raspberry Pi, Node-RED, InfluxDB, and Grafana in accordance with these specifications.

The research and requirement-gathering stage is an important part of developing the IOT based solar crop dryer with monitoring system. At this stage, the main focus is to understand how the dryer will be used and what information is required to monitor the drying process effectively. The requirements are identified by considering the operating conditions of the drying chamber, the parameters that can affect crop drying, and the way in which hardware and software components need to communicate with each other. The proposed system mainly focuses on monitoring important parameters such as temperature, humidity, and crop weight. These parameters provide useful information about the condition and progress of the drying process. For this reason, the system needs to collect sensor readings continuously at suitable time intervals. Since manual observation can be time-consuming, wireless communication is required to transfer the collected readings from the sensing unit to the processing system.

Based on the identified requirements, the ESP32 is selected for sensor data collection and wireless communication. MQTT is used to transfer the sensor information efficiently between the ESP32 and Raspberry Pi. The Local Computerprovides the main processing environment, while Node-RED is used to organize the data flow and perform required processing tasks.

Another important requirement is maintaining historical records of the drying process. InfluxDB is selected to store the continuously generated time-series sensor data. This stored information can later be used for analysis and comparison of different drying cycles. Finally, Grafana is used to provide a real-time dashboard where users can easily view temperature, humidity, weight, and drying progress. This requirement-gathering process helps ensure that the software architecture is

practical, organized, and capable of supporting future features such as prediction, automation, and remote monitoring.

5.1.1 Determining System Needs

In order to guaranty that the program can continuously receive sensor data, process the received data, store historical records, and give the user relevant information via a monitoring dashboard, the system requirements are determined. Throughout the design process, both functional and non-functional criteria are taken into account.

The system requirements are identified to ensure that the software can perform all the necessary tasks during the crop-drying process in a reliable and organized manner. The main requirement is that the system should be able to continuously receive sensor data from the drying chamber without unnecessary interruptions. The collected information should then be processed properly so that useful values can be presented to the user.

The system should also maintain historical records of important parameters such as temperature, humidity, and crop weight. Storing this information allows users to review previous drying cycles and understand how the drying conditions changed over time. A monitoring dashboard is also required to present the collected information in a simple and understandable format. Users should be able to view current readings as well as the progress of the drying process without needing to manually check each sensor.

Both functional and non-functional requirements are considered during the system design. Functional requirements describe what the system should do, such as data collection, communication, processing, storage, and visualization. Non-functional requirements focus on qualities such as reliability, usability, scalability, and maintainability. Considering both types of requirements helps create a system that is practical for current use and flexible enough for future improvements.

5.1.2 Identification of Monitoring Parameters

Crop weight, temperature, and humidity are thought to be crucial factors for tracking the drying process. The sensing system continuously gathers these characteristics, which are then sent to the software platform for additional processing and visualization.

Identifying the right monitoring parameters is an important step in designing the IOT based solar crop dryer with monitoring system. The drying process is affected by several environmental and product-related conditions, so selecting suitable parameters helps the system provide useful information about the actual condition of the crop. For the proposed system, crop weight, temperature, and humidity are considered the main parameters for continuous monitoring.

Temperature is important because it directly influences the rate at which moisture is removed from the crop. Monitoring the temperature inside the drying chamber helps the user understand whether the drying environment is suitable and whether the temperature is changing during the process. Humidity is also monitored because it indicates the amount of moisture present in the surrounding air. Changes in humidity can provide useful information about how moisture is being released from the crop during drying.

Crop weight is another important parameter because the weight of the crop generally decreases as moisture is removed. By observing changes in weight over time, the system can provide an indication of the drying progress. Combining weight information with temperature and humidity gives a more complete view of the drying process.

Table 5.1 Identification of monitoring parameters

Sr.no	Parameter	Measurement Unit	Sensor	Purpose in Drying Monitoring
1	Temperature	°C	Temperature Sensor	Monitors the thermal condition inside the drying chamber
2	Relative Humidity	% RH	Humidity Sensor	Indicates the moisture condition of the air inside the chamber
3	Crop weight	kg/g	Load cell + HX711	Tracks weight reduction caused by moisture removal
4	Timestamp	Date and time	ESP32	Identifies when each measurement was recorded
5	Drying status	Status	Software derived	Provides an indication of the current drying condition
6	Drying progress	Percentage	Software derived	Helps observe the progress of the drying cycle

The sensing unit continuously collects these parameters and sends the readings to the software platform through the ESP32 and MQTT communication system. The received data is then processed and stored for further use. The information can be displayed through the Grafana dashboard using graphs and numerical values, allowing users to easily observe changes during the drying cycle. Historical readings can also be reviewed to understand previous drying cycles and support future analysis. These parameters also provide a useful foundation for future features such as moisture estimation, drying-time prediction, automated alerts, and intelligent drying control.

5.2 System Architecture and Workflow

Each component of the suggested system's layered software architecture serves a distinct purpose in the monitoring process as a whole. The Local Computersupplies the central computing environment, and the ESP32 serves as the data gathering device. InfluxDB stores the gathered data, Grafana offers the user interface, Node-RED controls the data flow, and MQTT facilitates communication. The system architecture of the proposed IOT based solar crop dryer with monitoring system is designed as a layered software structure in which each component performs a specific task. The main objective of this architecture is to create a smooth flow of information from the sensors to the final monitoring dashboard. By separating data collection, communication, processing, storage, and visualization, the system becomes easier to understand, manage, and improve in the future.

The process starts with the ESP32, which acts as the primary data-gathering device. Sensors connected to the ESP32 continuously measure important drying parameters such as temperature, humidity, and crop weight. After collecting the readings, the ESP32 uses Wi-Fi to send the sensor data to the processing environment.

MQTT is used as the communication protocol between the ESP32 and Raspberry Pi. An MQTT broker manages the exchange of messages and ensures that the collected sensor information reaches the required software components. The Local Computeracts as the central computing platform where the received information is processed and managed.

Node-RED plays an important role in controlling the data flow. It receives the incoming sensor readings and connects different parts of the software system. It can also be used to apply basic processing, validation, and automation logic before the data is stored.

After processing, the sensor readings are stored in InfluxDB. Since the system continuously generates data over time, InfluxDB provides a suitable environment for maintaining time-based historical records. These records can later be used to analyze drying cycles and observe changes in the monitored parameters.

Finally, Grafana provides the user interface through which the collected information can be viewed. The dashboard can display real-time and historical values using graphs, numerical indicators, and other visual elements. This layered architecture provides a clear and organized data flow while also creating a suitable foundation for future features such as automated control, remote monitoring, drying-time prediction, and moisture forecasting.

5.2.1 Sensor Data Acquisition

The drying system's sensor readings are gathered by the ESP32, which then gets the data ready for transmission. Temperature, humidity, and crop weight data are among the obtained values. Data acquisition is the first and an important stage of the proposed IOT based solar crop dryer with monitoring system. At this stage, the main purpose is to collect useful information from the drying chamber and prepare it for further processing. The ESP32 acts as the main data acquisition unit and is connected to the sensors installed in the drying system. It collects sensor readings at regular intervals so that the drying conditions can be monitored continuously throughout the drying cycle.

The main parameters considered for data acquisition are temperature, humidity, and crop weight. The temperature sensor provides information about the conditions inside the drying chamber, while the humidity sensor measures the moisture level of the surrounding air. Crop weight is monitored using a load cell, which helps identify changes in the weight of the crop as moisture is removed during the drying process. These three parameters together provide useful information about the overall progress of drying.

After collecting the readings, the ESP32 organizes the sensor values into a suitable format before transmitting them to the processing system. The device uses its built-in Wi-Fi capability to establish wireless communication with the Raspberry Pi. MQTT is used for transferring the collected data because it is lightweight and suitable for continuous IoT communication.

The ESP32 can also perform basic checks before sending the readings, such as ensuring that the received values are available and within an expected range. This helps maintain better data quality during the monitoring process. Once the data is transmitted successfully, it is received by the Local Computer and passed to Node-RED for further processing and management. Thus, the data acquisition stage provides the basic information required for real-time monitoring, historical analysis, and future intelligent features of the drying system.

5.2.2 Communication Architecture

The MQTT communication protocol is used to send the gathered sensor data over Wi-Fi. The communication between the ESP32 and the processing environment is controlled by a MQTT broker.

Data communication is responsible for transferring the sensor information collected by the ESP32 to the processing environment. In the proposed IOT based solar crop dryer with monitoring system, Wi-Fi is used as the wireless connection, while MQTT is used as the main communication protocol. This combination allows the sensor readings to be transferred without requiring a physical connection between the ESP32 and Raspberry Pi.

After collecting parameters such as temperature, humidity, and crop weight, the ESP32 prepares the readings and publishes them through MQTT. An MQTT broker acts as an intermediate communication point between the ESP32 and the receiving software. It receives the messages from the ESP32 and forwards them to the appropriate subscriber, such as the Node-RED system running on the Localhost.

MQTT is suitable for the proposed system because it is lightweight and designed for communication between IoT devices. It can handle frequent sensor updates while using relatively low network resources. The use of a broker also keeps the sensing and processing components loosely connected, making it easier to add other devices or software services in the future.

Once the sensor data reaches the ESP32, it is made available to Node-RED for further processing, storage, and visualization. This communication layer therefore provides a reliable connection between the physical sensing unit and the software-based monitoring system.

5.2.3 Sensor Data Processing

Node-RED is utilized to process and arrange the incoming sensor data after the Local Computergets the transmitted data. After processing, the data is sent for storage and viewing.

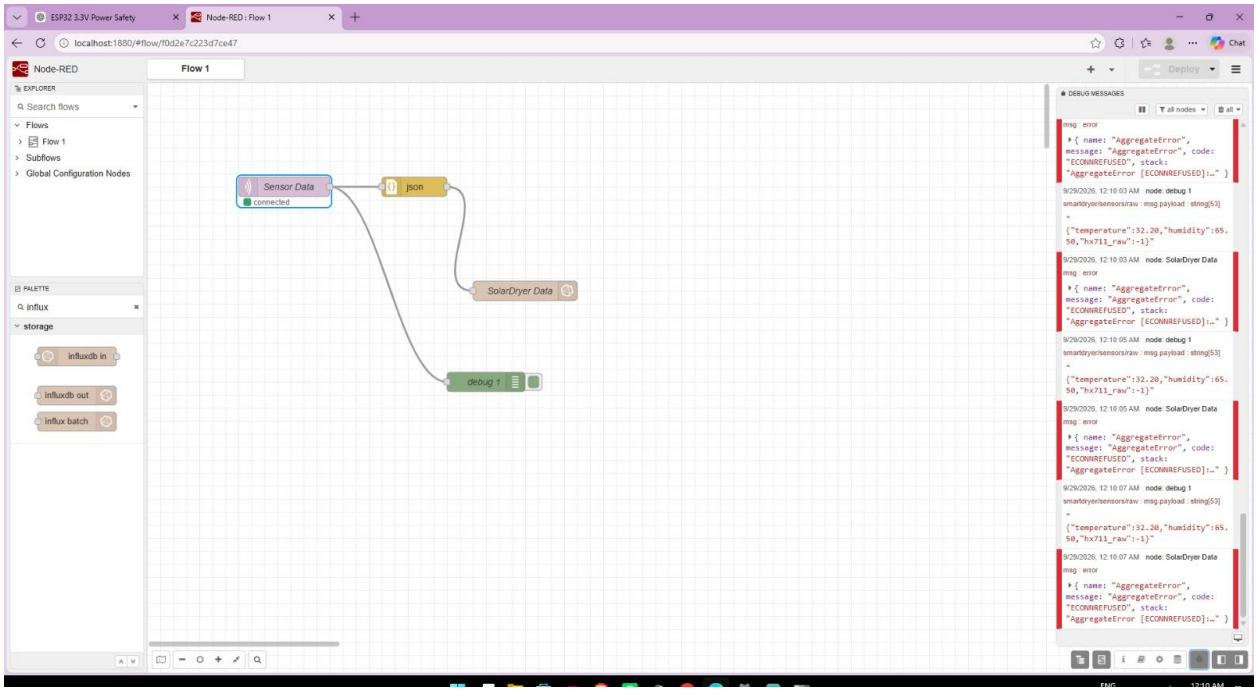


Fig 5.1 Shows Node-RED communication

Data processing is the stage where the sensor information received from the ESP32 is organized and prepared for storage and visualization. After the Local Computerreceives the sensor readings through MQTT, Node-RED is used to manage and process the incoming data. It acts as an important link between the communication layer, database, and monitoring dashboard. Node-RED receives values such as temperature, humidity, and crop weight and arranges them into a structured

format. Basic processing can be performed at this stage to check whether the received values are available and suitable for further use. The data can also be associated with timestamps so that each reading can be identified according to when it was recorded. This is useful for observing how the drying conditions change over time. Once the data has been processed, Node-RED forwards it to InfluxDB for historical storage. At the same time, the processed information can be made available to Grafana for real-time visualization. This allows the user to observe the current drying conditions through the monitoring dashboard.

The Node-RED workflow also provides flexibility for adding additional processing or automation logic in the future. For example, alerts, threshold-based conditions, drying-status calculations, or other data analysis functions can be incorporated without changing the complete system architecture. Thus, the data-processing stage helps convert raw sensor readings into useful information for monitoring and analysis.

5.2.4 Sensor Data Storage

The sensor measurements are stored as time-series data in InfluxDB. The technology is able to save previous data regarding the drying process because each reading is linked to a timestamp. Data storage is an important part of the proposed IOT based solar crop dryer with monitoring system because the sensor readings generated during the drying process need to be maintained for both current monitoring and future analysis. In the proposed system, InfluxDB is used to store the collected sensor measurements as time-series data. It is suitable for this application because the drying system continuously generates readings at different points in time.

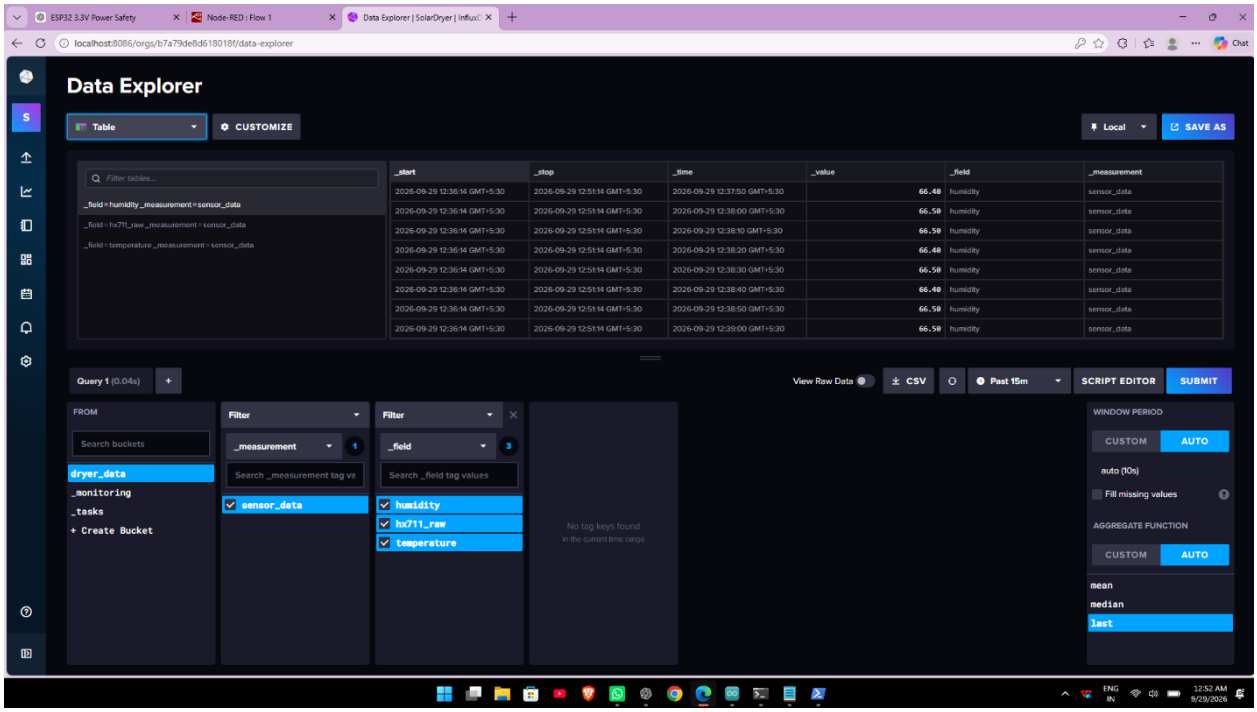


Fig 5.2 Shows Sensor data storage in InfluxDB

The data received from the ESP32 is processed through Node-RED before being stored in InfluxDB. Important parameters such as temperature, humidity, and crop weight are recorded along with their corresponding timestamps. The timestamp helps identify exactly when each measurement was taken and makes it possible to understand how the drying conditions changed throughout the drying cycle.

Storing the readings over time provides a useful historical record of the complete drying process. For example, users can review how the temperature and humidity changed while the crop weight gradually decreased. This information can help in understanding the relationship between environmental conditions and crop drying progress.

The stored data can also be accessed by Grafana to generate real-time and historical visualizations. Users can select a particular time period and observe previous readings through graphs, making it easier to compare different drying cycles. Historical records can also support future analysis and improvements to the system.

The database structure is designed to allow additional parameters to be added when required. Future data such as moisture level, energy consumption, airflow, or drying status could also be stored. Therefore, InfluxDB provides a flexible storage layer that supports the present monitoring requirements while creating a foundation for future prediction, analysis, and intelligent control features.

5.2.5 Data Visualization and Monitoring

Grafana offers a graphical monitoring interface through integration with InfluxDB. In an easy-to-understand manner, the dashboard displays temperature, humidity, crop weight, drying condition, and drying progress. Data visualization is the final stage of the monitoring process, where the stored sensor information is presented to the user in a simple and understandable form. Grafana is used as the main visualization tool and is connected with InfluxDB to access both current and previously recorded drying data.

The Grafana dashboard can display important parameters such as temperature, humidity, and crop weight using graphs, numerical values, and other visual indicators. This makes it easier for the user to understand the current condition of the drying chamber without manually checking the sensors.

The dashboard can also show changes in these parameters over time. For example, users can observe how the temperature and humidity change during the drying cycle and how the crop weight decreases as moisture is removed. This provides a clearer view of the overall drying progress.

Another useful feature is the ability to view historical data. Since the readings are stored in InfluxDB with timestamps, Grafana can display previous drying records and allow users to compare different stages or drying cycles. The dashboard can also include information about the current drying condition and progress based on the collected sensor data.

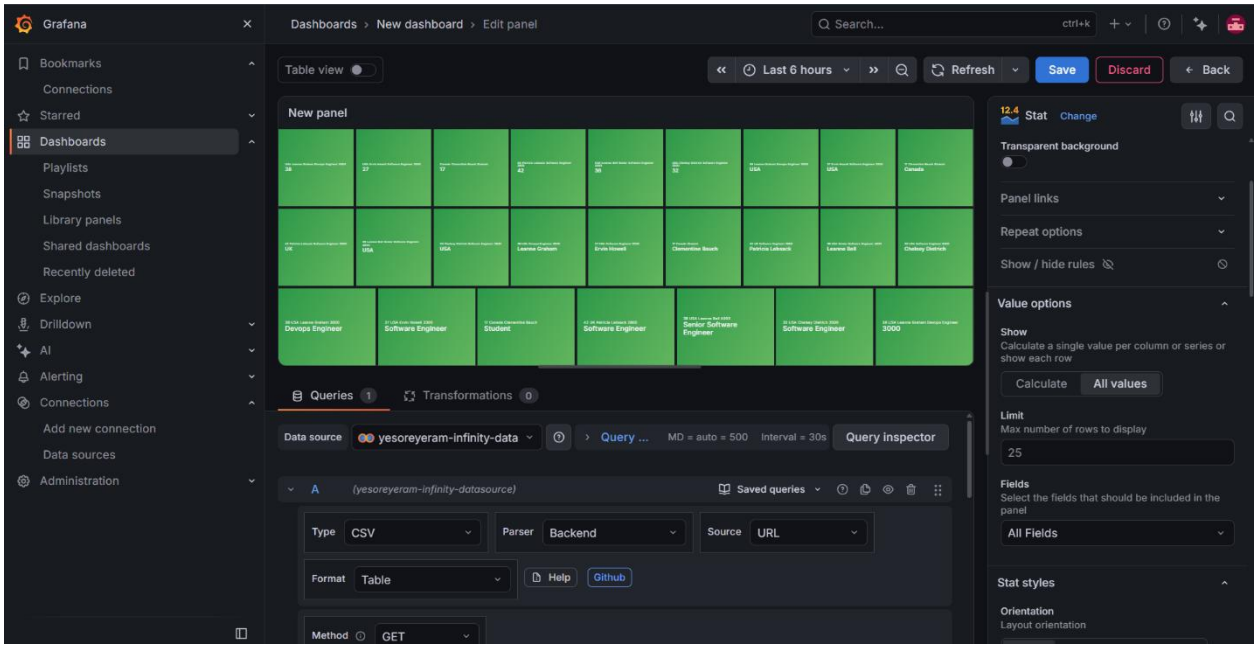


Fig 5.3 Shows Grafana Dashboard visualization

The visualization layer is designed to reduce the complexity of raw sensor data and convert it into information that can be easily understood. A clear dashboard can help users identify changes in drying conditions and take appropriate action when required. In the future, the same dashboard can be extended to include alerts, moisture predictions, drying-time estimation, and other intelligent monitoring features.

5.3 Software Design and User Interface

The purpose of the software interface is to give the user an easy way to watch the drying process without having to engage directly with each sensor. The dashboard uses the stored sensor data to display both historical and real-time statistics. The software design and user interface of the proposed IOT based solar crop dryer with monitoring system are developed with the main goal of making the monitoring process simple and convenient for the user. Instead of requiring the user to check individual sensors or physically inspect the drying chamber repeatedly, the software provides a centralized dashboard where important information can be viewed in one place.

The interface receives processed data from the system and presents it in a clear visual format. Important parameters such as temperature, humidity, crop weight, drying condition, and drying progress can be displayed using numerical values, graphs, and status indicators. This allows the

user to quickly understand the current condition of the drying chamber and observe how the parameters are changing over time.

The dashboard also makes use of historical data stored in InfluxDB. Users can view previous readings and examine the changes that occurred during earlier drying cycles. This historical information can be useful for comparing different drying sessions and understanding the relationship between environmental conditions and crop weight reduction.

The interface is designed to avoid unnecessary complexity and provide only the information that is relevant to the drying process. A well-organized dashboard can reduce the effort required for manual observation and help users follow the drying process more comfortably. The software architecture also allows the interface to be improved as new requirements are introduced.

In future development, additional features such as warning notifications, moisture-level estimation, drying-time prediction, remote access, and automated control indicators can be integrated into the same interface. Thus, the software design provides a practical connection between the technical data generated by the dryer and the information required by the user.

5.3.1 System Dashboard Design

The key drying parameters are arranged into a single monitoring interface using the Grafana dashboard. To give a clear picture of the current state of the system, numerical values and graphical representations are utilized.

Table 5.2 Dashboard Parameters and visualization

Sr.no	Dashboard Element	Data Source	Visualization method	Purpose
1	Temperature	InfluxDB	Line graph	Monitor chamber temperature
2	Humidity	InfluxDB	Line graph	Monitor relative humidity
3	Crop Weight	InfluxDB	Line graph	Observe weight reduction
4	Drying status	Processed sensor data	Status indicator	Indicate current drying condition
5	Drying progress	Processed data	Trend	Monitor drying progress
6	Historical trends	InfluxDB	Time Series graphs	Analyze previous drying cycles
7	Time information	timestamp	Time axis	Relate readings to drying duration

The key drying parameters are arranged into a single monitoring interface using the Grafana dashboard. To give a clear picture of the current state of the system, numerical values and graphical representations are utilized. The dashboard mainly focuses on important parameters such as temperature, humidity, and crop weight, which provide useful information about the progress of the drying process. Numerical values allow the user to quickly understand the present condition, while graphs help in observing how these parameters change over time.

The dashboard is designed to keep the information simple and easy to understand. Instead of checking individual sensors manually, the user can view the collected information in one place. This makes regular monitoring more convenient and reduces the need for continuous physical observation of the dryer. The graphical representation also makes it easier to identify changes or unusual variations in the drying conditions.

Historical data stored in InfluxDB can also be displayed through Grafana, allowing the user to review previous readings and compare them with the current drying cycle. This can help in understanding the overall drying pattern and observing how temperature, humidity, and crop weight change during the process. The dashboard therefore acts as a central interface between the user and the monitoring system.

In the future, the dashboard can be extended with additional features such as drying-status indicators, alerts, moisture estimation, drying-time prediction, and remote monitoring. This makes the dashboard flexible for future improvements and intelligent analysis.

5.3.2 Real-Time Monitoring

The user may track variations in temperature, humidity, and crop weight during the drying cycle by viewing the dashboard's regularly updated sensor information.

The user may track variations in temperature, humidity, and crop weight during the drying cycle by viewing the dashboard's regularly updated sensor information. The real-time monitoring feature helps the user understand the current condition of the dryer without checking each sensor separately. The ESP32 continuously collects readings from the connected sensors and sends the data through Wi-Fi using the MQTT communication protocol. The received information is processed through Node-RED and made available on the Grafana dashboard.

Temperature and humidity values help the user observe the environmental conditions inside the drying system, while crop weight provides an indication of the moisture being removed from the crop. As the drying process continues, changes in these values can be observed through numerical readings and graphical trends. This makes it easier to identify whether the drying conditions are changing as expected.

Real-time monitoring also reduces the need for frequent manual observation and makes the drying process more convenient to supervise. The continuously updated information can support timely decisions during operation. In future development, this feature can also be extended with alerts, drying-status indicators, and automatic control functions based on predefined conditions.

5.3.3 Drying Status and Progress

Information on the drying process's current state and advancement is provided by the program. This enables the user to track the drying process over time and determine whether it is still happening. Information on the drying process's current state and advancement is provided by the program. This enables the user to track the drying process over time and determine whether it is still happening. The drying status can be understood by observing the continuously updated sensor values displayed on the Grafana dashboard. Changes in temperature, humidity, and crop weight provide useful indications of how the drying process is progressing.

During the drying cycle, the crop weight generally changes as moisture is removed from the crop. At the same time, temperature and humidity readings help the user understand the conditions within the dryer. By observing these values together, the user can get a better idea of the current drying condition instead of depending only on manual checking.

The dashboard also provides a simple way to follow the progress of the drying cycle from its beginning to the later stages. Historical data stored in InfluxDB can be used to review previous readings and understand how the parameters changed over time. This makes the drying process easier to monitor, compare, and manage. In future, the system can include automatic drying-status indicators and alerts to provide more convenient monitoring.

5.3.4 Historical Data Visualization

Grafana graphs are used to display historical sensor data kept in InfluxDB. This allows users to examine how the system behaves at various phases of the drying process and analyze changes in drying parameters. Grafana graphs are used to display historical sensor data kept in InfluxDB. This allows users to examine how the system behaves at various phases of the drying process and analyze changes in drying parameters. Historical data visualization is useful because it gives the user a clear view of how the drying conditions have changed from the beginning to the end of a drying cycle.

The stored temperature, humidity, and crop weight readings can be represented using time-based graphs. By looking at these graphs, the user can easily identify increases, decreases, or other changes in the monitored parameters. For example, changes in crop weight over time can help the

user understand the progress of moisture removal, while temperature and humidity trends provide information about the conditions during drying.

The historical graphs also make it possible to compare different stages of the drying process. Instead of depending only on the current sensor readings, the user can look at previous values and understand the overall pattern. This can be helpful when reviewing completed drying cycles and identifying conditions that may have affected the drying process.

The combination of InfluxDB and Grafana provides an organized way to store and visualize this information. In the future, the collected historical data can also be used for further analysis, drying-time estimation, moisture prediction, and improving the overall efficiency of the IOT based solar crop dryer with monitoring system.

5.4 Integration and Implementation

The software and communication components are integrated into a single monitoring system during the implementation phase. A constant flow of data from the sensors to the user dashboard is established by the configuration of the ESP32, MQTT, Raspberry Pi, Node-RED, InfluxDB, and Grafana components.

The software and communication components are integrated into a single monitoring system during the implementation phase. A constant flow of data from the sensors to the user dashboard is established by the configuration of the ESP32, MQTT, Raspberry Pi, Node-RED, InfluxDB, and Grafana components. Each component performs a specific task, and their proper integration is important for the smooth operation of the overall system.

The ESP32 acts as the main data collection unit. It receives readings from the temperature, humidity, and crop weight sensors at regular intervals. The collected information is then transmitted through Wi-Fi using the MQTT communication protocol. The MQTT broker manages the messages and makes the sensor data available to the Node-RED.

Node-RED is used to manage the incoming data flow. It receives the MQTT messages, processes the sensor readings, and forwards the required information to InfluxDB for storage. InfluxDB stores the readings along with their corresponding timestamps, allowing the system to maintain a record of the drying process.

Grafana is connected to InfluxDB and uses the stored information to create an easy-to-understand monitoring dashboard. The user can view current sensor values as well as historical trends through numerical values and graphs. This integration creates a continuous data path from sensing to visualization and reduces the need for manual monitoring. The modular design also allows

additional sensors, analysis features, alerts, and intelligent control functions to be added in future development.

5.4.1 ESP32 and Sensor Integration

The ESP32, which is programmed to gather and prepare the sensor readings for transmission, is connected to the sensors. The ESP32, which is programmed to gather and prepare the sensor readings for transmission, is connected to the sensors installed in the drying system. It acts as the main interface between the physical sensors and the software monitoring platform. The sensors continuously provide information about important drying parameters such as temperature, humidity, and crop weight.

The ESP32 reads these values at regular intervals and prepares the collected information in a suitable format for further communication. The temperature and humidity readings provide information about the conditions inside the drying chamber, while the crop weight reading helps in observing the reduction in weight as moisture is removed during drying.

After collecting the sensor values, the ESP32 uses its built-in Wi-Fi capability to transmit the information to the processing environment. MQTT is used for communication so that the readings can be transferred efficiently to the Local Computer through the MQTT broker. The ESP32 can also perform basic checks on the collected values before transmission to help maintain consistent data.

This integration provides a direct connection between the physical drying environment and the software system. It allows sensor readings to be collected continuously without requiring manual recording. The collected data can then be processed, stored, and displayed through the remaining components of the system. The ESP32-based approach also provides flexibility for adding additional sensors in the future if more drying parameters need to be monitored.

5.4.2 MQTT and Local Computer Integration

The ESP32's sensor data is transferred to the Local Computer using the MQTT communication layer. After receiving the data via the MQTT broker, the Local Computer makes it available for additional processing. The MQTT and Local Computer integration provides the communication link between the ESP32-based sensing unit and the main processing environment. After collecting temperature, humidity, and crop weight readings, the ESP32 prepares the sensor information and sends it through Wi-Fi using the MQTT protocol. This allows the sensor data to reach the Local Computer without requiring a direct wired connection.

An MQTT broker is used to manage the communication between the devices. The ESP32 works as the publisher by sending sensor readings to specific MQTT topics. The Raspberry Pi, through

the required software services, subscribes to these topics and receives the published information. This publisher-subscriber approach keeps the sensing and processing components separated and makes the communication structure easier to manage.

Once the Local Computer receives the sensor data, the information is made available to Node-RED for further processing. Node-RED can organize the incoming messages, check the received values, and forward the processed information to the appropriate storage and visualization components. The Local Computer therefore acts as an important central point in the system where incoming sensor information is received and managed.

Using MQTT also makes the system flexible because additional sensors or devices can be connected later without completely changing the existing architecture. The communication layer can handle regular sensor updates while keeping the data flow organized. This integration creates a continuous path from the sensors to the processing system and supports real-time monitoring of the drying process. It also provides a suitable foundation for future features such as alerts, automated control, remote monitoring, and data-based analysis.

5.4.3 Node-RED and Database Integration

After processing the incoming sensor data, Node-RED sends the necessary information to InfluxDB for storage. This creates the link between historical data management and data acquisition.

After receiving the sensor information from the ESP32 through MQTT, Node-RED acts as an important processing layer between the data acquisition system and the database. Its main purpose is to organize the incoming sensor readings and prepare them for storage in InfluxDB. This integration ensures that the data collected during the drying process is not only available for real-time monitoring but is also maintained for future reference.

Node-RED receives parameters such as temperature, humidity, and crop weight from the MQTT communication flow. The incoming values can be checked and arranged into a structured format before they are sent to the database. A timestamp is associated with each reading so that the stored information can be related to a particular point in the drying cycle. This is especially useful for understanding how the drying conditions change over time.

Once the data is properly prepared, Node-RED forwards it to InfluxDB, where it is stored as time-series data. InfluxDB maintains the historical readings and allows them to be retrieved whenever required. The stored information can later be used to analyze completed drying cycles and compare changes in temperature, humidity, and crop weight.

The connection between Node-RED and InfluxDB also keeps the system organized because data processing and data storage are handled as separate stages. If additional parameters or processing requirements are introduced in the future, the Node-RED workflow can be modified without redesigning the entire system. This integration therefore provides a continuous data path from sensor acquisition to historical storage and creates a strong foundation for future analysis, prediction, and intelligent control features.

5.4.4 Grafana Integration

In order to retrieve the stored sensor data and present it via the monitoring dashboard, Grafana is linked to the InfluxDB database.

Grafana is integrated with the InfluxDB database to provide a graphical interface for monitoring the drying process. The main purpose of this integration is to convert the stored sensor readings into information that can be easily understood by the user. Since InfluxDB maintains the sensor measurements along with their timestamps, Grafana can retrieve both recent and historical data and display it through an interactive dashboard.

The dashboard is designed to present important drying parameters such as temperature, humidity, and crop weight. Numerical values can be used to show the latest sensor readings, while graphs can represent how these parameters change over time. This makes it easier for the user to observe the behaviour of the drying chamber without directly checking the sensors or manually recording their readings.

Grafana can also display historical information stored in InfluxDB. For example, the user can select a particular time period and observe the changes in temperature, humidity, and crop weight during that period. This can help in understanding the different stages of the drying cycle and reviewing the performance of the system after a drying operation has been completed.

The dashboard can be organized into separate panels so that each important parameter is clearly visible. Additional information such as drying status and progress can also be presented based on the available sensor data. The visual representation of the information makes it easier to identify changes, trends, or unusual variations in the drying conditions.

The Grafana integration also supports future improvements to the system. Additional panels can be added for parameters such as moisture level, energy consumption, or airflow if the required sensors are introduced. Alerts and other monitoring features can also be incorporated later. Therefore, the connection between Grafana and InfluxDB provides the final visualization layer of the system and creates a convenient interface between the collected data and the user.

5.4.5 System Integration and Testing

The entire data flow, from sensor acquisition to dashboard visualization, is tested after all components have been integrated. The system's correct data transfer, storage, visualization, and ongoing operation are all examined. Before final implementation, any issues found are fixed.

The entire data flow, from sensor acquisition to dashboard visualization, is tested after all components have been integrated. The main purpose of testing is to make sure that every part of the IOT based solar crop dryer with monitoring system works correctly both individually and as part of the complete system. The testing process focuses on checking whether sensor readings are collected properly, transferred without unnecessary interruption, processed correctly, stored in the database, and displayed accurately on the Grafana dashboard.

First, the connection between the sensors and ESP32 is checked to confirm that temperature, humidity, and crop weight readings are being received correctly. The communication between the ESP32 and Node-RED is then tested using Wi-Fi and MQTT. The MQTT broker is observed to ensure that messages are being published and received properly.

The Node-RED workflow is also tested to verify that incoming sensor data is processed and forwarded to the correct destination. The storage connection with InfluxDB is checked to make sure that readings are saved with the appropriate timestamps. This is important because the historical data will later be used for analysis and visualization.

The Grafana dashboard is tested to confirm that the displayed values correspond to the data stored in InfluxDB. Both real-time readings and historical graphs are checked for proper updates and accurate representation. The system is also observed over a continuous period to identify communication interruptions, missing readings, or unexpected behavior.

Any errors found during testing are analyzed and corrected before the final implementation. This testing process helps ensure that the complete software pipeline operates smoothly and provides reliable information to the user during the drying process.

CHAPTER – VI

RESULT AND DISCUSSION

CHAPTER - VI

RESULT AND DISCUSSION

6.1 Introduction

This chapter presents the results of the software developed for the IOT based solar crop dryer with monitoring system. The results are discussed in terms of the functional performance of each software layer, the working of the complete data pipeline from sensors to dashboard and the extent to which the objectives listed in Section 1.3 have been achieved. The implemented pipeline consists of sensor acquisition through the ESP32, MQTT-based communication, data processing in Node-RED, time series storage in Influx DB and visualization through Grafana.

6.2 Implementation Results

The software was implemented as a modular pipeline in which each stage passes its output to the next. The outcome of each stage is summarized in table 6.1.

Table 6.1 Summary of implementation results

Sr.no	Module	Function	Result
1	ESP32 firmware (Arduino IDE)	Reads temperature, humidity and weight; connects to Wi-Fi; publishes over MQTT	Sensor readings are acquired at regular intervals and published to defined MQTT topics
2	MQTT Broker	Manages message exchange between ESP32 and Local Computer	Published readings are received and forwarded to the subscribing Node-RED flow
3	Node-RED	Receives, validates, timestamps and routes data	Incoming readings are structured, timestamped and forwarded to Influx DB
4	Influx DB	Time-series storage	Temperature, humidity and weight are stored with timestamps as historical records
5	Grafana	Real-time and historical dashboard	Values are displayed as numerical indicators and time-series graphs

6.2.1 Data Acquisition and communication

The ESP32 reads temperature, humidity and crop weight (load cell with HX711) at regular intervals and publishes them through the MQTT broker over Wi-Fi. The publish-subscribe model kept the sensing unit and the processing unit loosely coupled, so the data flow continued without the ESP32 needing to communicate directly with each software component.

6.2.2 Data processing and storage

The Node-RED flow receives the MQTT messages, arranges them into a structured format, applies basic validation and timestamping, and forwards them to Influx DB. The reading are stored as time-series records which preserves the complete history of each drying cycle and allows any earlier period to be retrieved later.

6.2.3 Dashboard Visualization

The Grafana dashboard retrieves the stored data from Influx DB and presents temperature, humidity and crop weight as numerical values and time-based graphs. The time range selector allows the user to move between the current readings and earlier records without any manual data handling.

6.3 Functional Verification

The complete data path was checked stage by stage, as described in section 5.4.5. The verification results are given in Table 7.2.

Table 6.2 Functional verification of the system

Sr.no	Verification point	Outcome
1	Sensor readings available at the ESP32	Verified
2	Messages published to the MQTT broker	Verified
3	Messages received and processed by Node-RED	Verified
4	Readings written to Influx DB with timestamps	Verified
5	Grafana panels reflect the data stored in Influx DB	Verified

6	Historical data retrievable by time range	Verified

These checks confirm that the software supports a continuous flow of data from the sensing unit to the user dashboard which was the central requirement of the system.

6.4 Discussion

Real-time monitoring: The dashboard lets the user view the drying condition from one location, without the drying chamber or reading individual sensors. This directly addresses the dependence on manual supervision identified in the problem statement.

Weight as a progress indication: Since crop weight decreases as moisture is removed, the weight graph a direct indication of drying indication of drying progress. Read together with the temperature and humidity graphs, it shows how chamber conditions relate to the rate of moisture loss.

Historical records: Storing every reading in Influx DB creates a record of each drying cycle that traditional methods do not provide. Completed cycles can be reviewed and compared and this data is the basis for the prediction features proposed as future work.

Modularity: Because communication, processing, storage and visualization are separate components each could be configured and check independently. Sensors, alerts or analysis function can be added later without redesigning the system.

Comparison with existing work: The systems reviewed in chapter II largely focus on individual functions such as sensor monitoring or remote display. The developed software combines acquisition, communication, processing, storage and visualization in one framework, which addresses the research gap identified in section 2.3.

6.5 Achievement of objectives

Table 6.3 Objectives and outcomes

Sr.no	Objective	Outcome
1	Real-time monitoring of temperature, humidity and weight	Achieved through the Grafana dashboard
2	Sensor data transmission using ESP32 and MQTT	Achieved
3	Data processing using Local Computer and Node-RED	Achieved
4	Historical storage in Influx DB	Achieved

5	Interactive Grafana dashboard	Achieved
6	Reduced manual supervision	Achieved through automated logging and visualization
7	Foundation for future features	Provided by the modular architecture and stored data

6.6 Limitations

- The results in this chapter are functional. Quantitative accuracy analysis of the sensors and load-cell calibration against reference instruments is left for further work.
- Drying status and progress indicators are based on simple sensor-derived logic and need tuning with data from repeated drying cycles of different crops.
- The system depends on a stable Wi-Fi connection and a continuously running Local Computer.
- Moisture prediction, drying-time estimation and automated control are proposed as future work (Section 6.2) and are not part of the present implementation.

6.7 Summary

The developed software provides a working pipeline that acquires sensor data with the ESP32, transmits it over MQTT, processes it in Node-RED, stores it in Influx DB and visualizes it in Grafana. The results show that the objectives of continuous monitoring, historical storage and dashboard visualization have been met, and the modular design provides a base for prediction, alerts and automated control.

CHAPTER – VII

CONCLUSION AND SUGGESTED FURTHER WORK

CHAPTER - VII

CONCLUSION AND SUGGESTED FURTHER WORK

7.1 Conclusion

An integrated platform for tracking and controlling the crop-drying process is offered by the suggested software system for the IOT based solar crop dryer with monitoring system. To offer constant information regarding the drying conditions, the system integrates sensor-based data collecting, wireless transmission, data processing, storage, and visualization.

Important characteristics including temperature, humidity, and crop weight are gathered by the ESP32, which then uses MQTT to send the data via Wi-Fi. Node-RED controls the processing and data flow, while the Local Computer supplies the primary computing environment. Grafana offers a graphical user interface for both historical and real-time monitoring, while InfluxDB keeps track of past sensor data.

The suggested architecture offers an organized technique for drying process analysis and lessens reliance on ongoing manual observation. The software system can be expanded in the future with more sophisticated and analytical functions thanks to the modular design. All things considered, the suggested solution offers a good software base for creating a data-driven, Internet of Things-enabled multi-crop drying system.

7.2 Suggested Further Work

The proposed system can be further enhanced in several areas to improve its functionality and intelligence:

1. Moisture Prediction – Machine-learning models can be incorporated to estimate the moisture content of crops during the drying process.
2. Drying-Time Prediction: The remaining drying time for various crops and operating conditions can be predicted by analyzing historical sensor data.
3. Automated Drying Control: Upcoming iterations will be able to use sensor data to autonomously regulate the drying process based on predetermined parameters.
4. Remote Monitoring: Using an internet-connected application, the dashboard may be expanded to let customers keep an eye on the drying process from a distance.
5. Multi-Crop Analysis: To determine each crop's unique drying patterns and needs, the system can be taught using data from several crops.

6. Machine Learning Integration: Predictive models can be created and decision-making enhanced by using historical temperature, humidity, weight, and drying data.
7. Data Analytics: To compare various drying cycles, spot trends, and assess system performance, advanced analysis can be implemented.
8. Mobile Application: To give consumers notifications, real-time monitoring, and drying-status information, a specific mobile application might be created.

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